

start : program
;

program : program unit
| unit
;

unit : variable_decl
| func_definition
;

func_definition : type_specifier ID LPAREN param_list RPAREN compound_statement
| type_specifier ID LPAREN RPAREN compound_statement
;

param_list : param_list COMMA type_specifier ID
| param_list COMMA type_specifier
| type_specifier ID
| type_specifier
;

compound_statement : LCURL statements RCURL
| LCURL RCURL
;

variable_decl : type_specifier declaration_list SEMICOLON
;

type_specifier : INT
| FLOAT
| VOID
| CHAR
;

declaration_list : declaration_list COMMA ID
| declaration_list COMMA ID LTHIRD CONST_INT RTHIRD
| ID
| ID LTHIRD CONST_INT RTHIRD
;

statements : statement
| statements statement
;

statement : variable_decl
| **expression_statement**
| **compound_statement**
| **FOR LPAREN expression_statement expression_statement expression RPAREN**
| **statement**
| **IF LPAREN expression RPAREN statement**
| **IF LPAREN expression RPAREN statement ELSE statement**
| **WHILE LPAREN expression RPAREN statement**
| **PRINTLN LPAREN ID RPAREN SEMICOLON**
| **RETURN expression SEMICOLON**
;

expression_statement : SEMICOLON
| **expression SEMICOLON**
;

variable : ID
| **ID LTHIRD expression RTHIRD**
;

expression : logic_expression
| **variable ASSIGNOP logic_expression**
;

logic_expression : rel_expression
| **rel_expression LOGICOP rel_expression**
;

rel_expression : simple_expression
| **simple_expression RELOP simple_expression**
;

simple_expression : term
| **simple_expression ADDOP term**
;

term : unary_expression
| **term MULOP unary_expression**
;

unary_expression : ADDOP unary_expression
| **NOT unary_expression**
| **factor_info**
;

factor_info : factor
;

factor : variable

- | ID LPAREN argument_list RPAREN
- | LPAREN expression RPAREN
- | CONST_INT
- | CONST_FLOAT
- | variable INCOP
- | variable DECOP

;

argument_list : arguments

|
;

arguments : arguments COMMA logic_expression

| logic_expression
;